

Model Evaluation Metrics — KS Score

What is KS Score? How's it computed and used?



Tarun_KS · Follow

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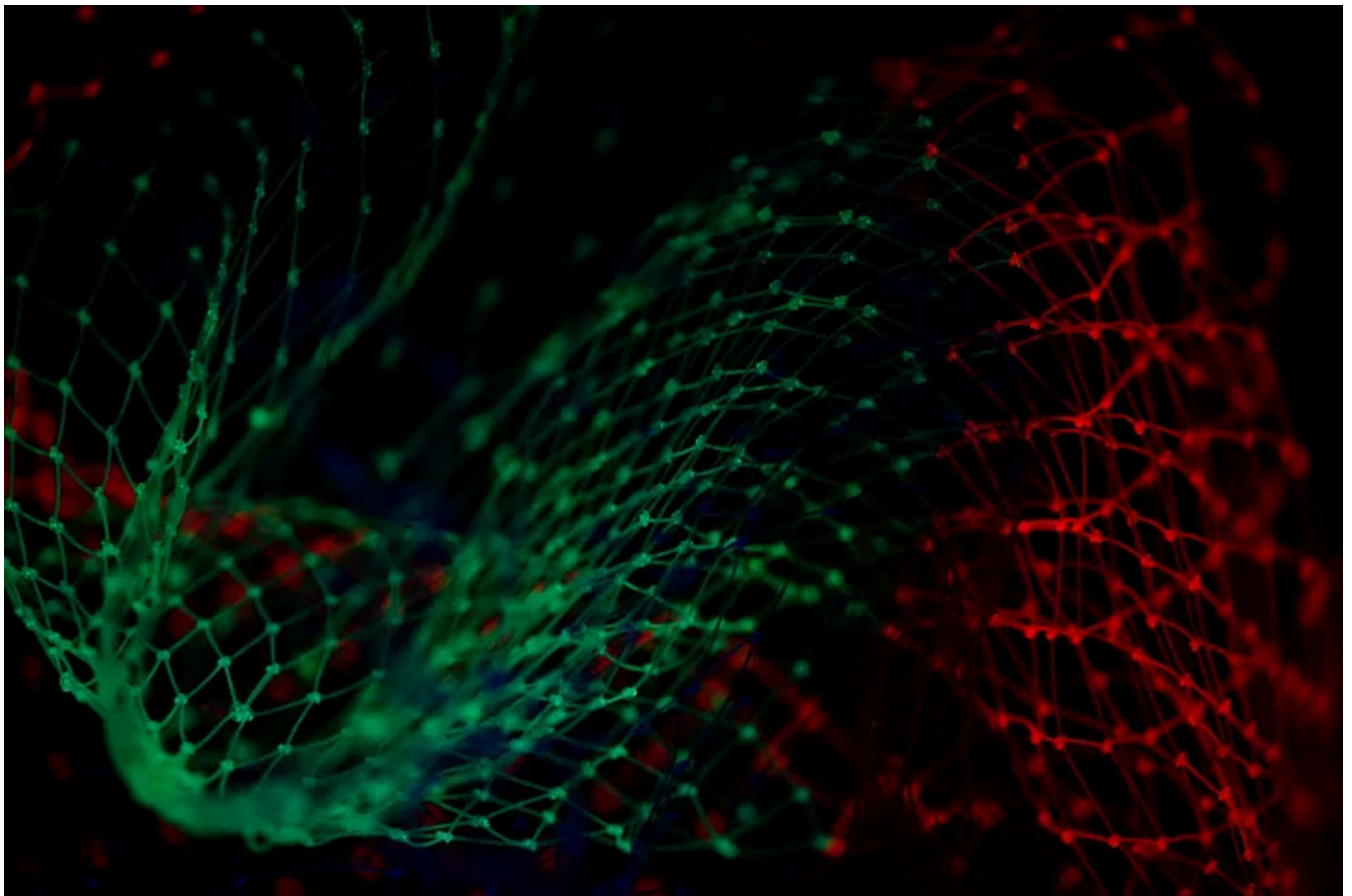


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What are evaluation metrics & why do we need to evaluate a model?

- *Evaluation metrics are those which are used to quantify the performance of an ML*

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trained by providing a score.

- *Model evaluation helps to identify problems with the model and the data. It provides an opportunity to optimize the model, for example, by tuning hyperparameters or selecting different features.*
- *Finally, quantifying the model performance will build trust in the model and the stakeholders who rely on the model for decision-making will be able to get a measure of the model's predictive power.*

What is KS Score?

- **Kolmogorov-Smirnov (KS) score** is a way of comparing the cumulative sum of the positive and negative classes. It measures the maximum difference between the two over the range of predicted probabilities.
- The population of the entire dataset is divided into deciles based on the prediction probability. Each decile is generally known as a **Tier**.
- A high KS score indicates that the model has a better separation between the positive (goods) and negative classes (bads) at the respective tier. This metric is used widely in the finance and credit risk industry.
- KS Score provides a single scalar value that is easy to interpret and compare across different models.

Ideally, the max separation should be present in the riskiest 3 tiers (Top 3 tiers if DV = 0 is bad or bottom 3 tiers if DV = 1 is bad).

Computing KS Score

```
class ModelEvaluation:

    """
    A class to compute the gains table for a given predictions DataFrame.

    Attributes
    -----
    predictions : pd.DataFrame
        A DataFrame containing 'PD', 'DV' columns.
```

Methods

```

-----
compute_gains_table_ks():
    Computes the gains table from the provided predictions DataFrame.
    Returns the gains table & KS value from the gains table.

"""

def __init__(self, predictions_df):
    self.predictions = predictions_df

def compute_gains_table_ks(self):

    self.predictions['Bins'] = pd.qcut(self.predictions['PD'],q = 10)
    self.predictions['Total'] = 1
    self.predictions.rename(columns={'DV':'Bads'},inplace=True)

    gains_table = self.predictions[['Bins', 'Bads', 'Total']].groupby(self.

    gains_table['Goods'] = gains_table['Total'] - gains_table['Bads']
    gains_table['Cumulative_bads'] = gains_table['Bads'].cumsum()
    gains_table['Cumulative_goods'] = gains_table['Goods'].cumsum()
    gains_table['Cumulative_Bad_Rate'] = (gains_table['Cumulative_bads'] /
    gains_table['Cumulative_Good_Rate'] = (gains_table['Cumulative_goods']
    gains_table['KS'] = abs(gains_table["Cumulative_Bad_Rate"] - gains_table["Cumulative_Good_Rate"])
    gains_table = gains_table[['Bads', 'Goods', 'Total', 'Cumulative_bads', 'Cumulative_goods', 'KS']]
    model_ks = gains_table['KS'].max()

def identify_max(series):
    max_value = series.max()
    return ['<---' if i == max_value else '' for i in series]

def highlight_max(series):
    max_value = series.max()
    return ['background-color: aquamarine' if i == max_value else '' for i in series]

# Apply the identify_max & highlight_max functions to the DataFrame
gains_table['Max_KS'] = identify_max(gains_table['KS'])
gains_table = gains_table.style.apply(highlight_max,subset=['KS'])

return gains_table, model_ks

```

	PD	DV
0	0.107073	0
1	0.329556	0
2	0.163740	0
3	0.304674	1
4	0.149844	1

Sample predictions data frame (where Label 0 — Class Good & Label 1 — Class Bad)

```
model_score = ModelEvaluation(predictions_df)
gains_table, ks = model_score.compute_gains_table_ks()
```

	Bads	Goods	Total	Cumulative_bads	Cumulative_goods	Cumulative_Bad_Rate	Cumulative_Good_Rate	KS	Max_KS
Bins									
(0.028, 0.0565]	37	963	1000	37	963	1.650000	12.400000	10.750000	
(0.0565, 0.0937]	81	919	1000	118	1882	5.280000	24.240000	18.960000	
(0.0937, 0.125]	102	898	1000	220	2780	9.840000	35.810000	25.970000	
(0.125, 0.158]	158	842	1000	378	3622	16.910000	46.650000	29.740000	
(0.158, 0.194]	190	810	1000	568	4432	25.400000	57.080000	31.680000	
(0.194, 0.237]	189	811	1000	757	5243	33.860000	67.530000	33.670000	<---
(0.237, 0.289]	253	747	1000	1010	5990	45.170000	77.150000	31.980000	
(0.289, 0.352]	309	691	1000	1319	6681	58.990000	86.050000	27.060000	
(0.352, 0.452]	388	612	1000	1707	7293	76.340000	93.930000	17.590000	
(0.452, 0.772]	529	471	1000	2236	7764	100.000000	100.000000	0.000000	

Gains Table

We can see that the KS Score for this dataset based on the predictions obtained from the ML model is 33.69.

Conclusion

To summarise, the KS Score serves as a powerful tool for quantifying the discriminatory power of predictive models, particularly in the realm of binary classification problems. *KS Score is very helpful in identifying the separation between goods and bads at each tier defined by the probability ranges.*

As a data scientist, it is crucial to understand and effectively utilize various model evaluation metrics, including the KS Score, to ensure that the models are not only accurate but also robust and reliable in practice. *By incorporating the KS Score into one's evaluation toolkit, we can make more informed decisions about model selection, refinement, and deployment, ultimately driving better outcomes for the organization and stakeholders.*

Thanks for reading the blog. If there are any questions or if you want to share any valuable feedback, feel free to reach out to me on [Linkedin](#).

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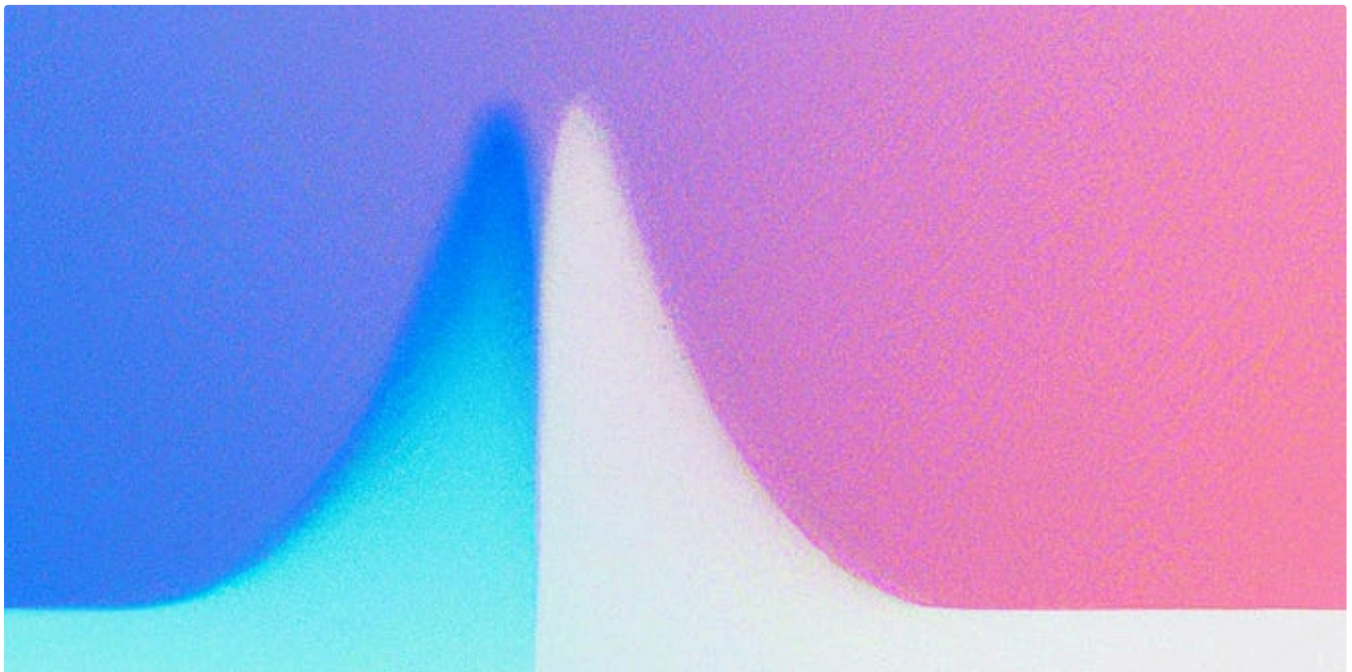


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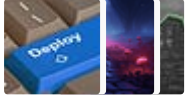
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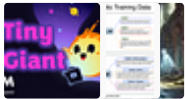
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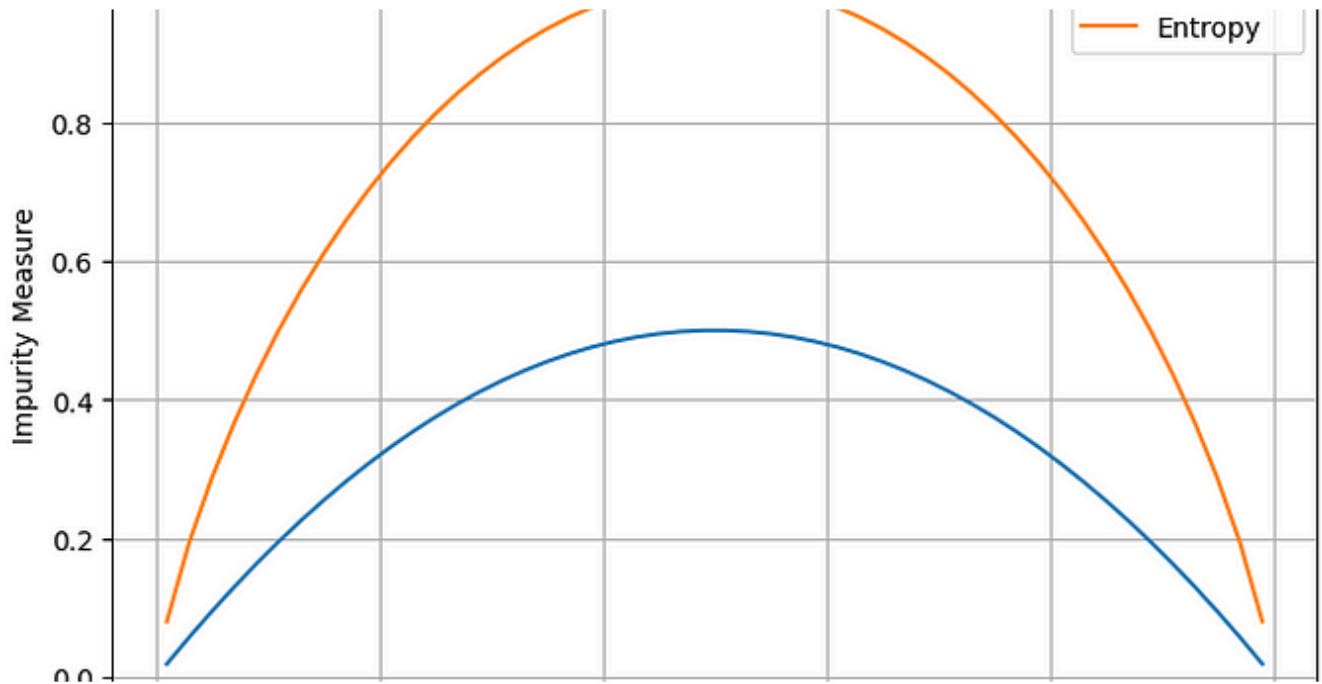
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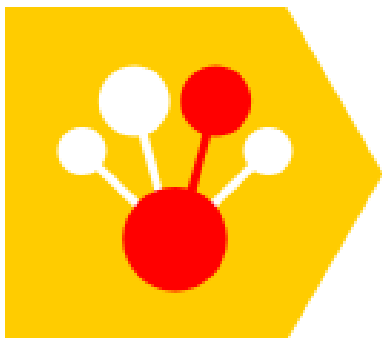


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